

MUHAMMAD IRFAN BIN RIZAL

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SUMMARY

Cloud Computing & Application student at Universiti Teknikal Malaysia Melaka with hands-on expertise in AWS cloud security, network infrastructure, and full-stack development. Holder of AWS Academy Cloud Security Builder, Oracle Cloud Infrastructure/Fusion AI Agent Studio, and multiple Cisco cybersecurity credentials. 12+ end-to-end projects spanning privacy-first messaging, autonomous cloud security response, multi-cloud disaster recovery, zero-trust data governance, and CSPM automation. Gold Medalist at i-CPROM 2023 (UiTM Perlis) for IT innovation, with a 6-month IT internship at Port Klang Cruise Terminal and an active faculty technical support role.

EDUCATION

Universiti Teknikal Malaysia Melaka (UTeM) 2024 – Present
Bachelor of Technology in Cloud Computing and Application (Honours) • CGPA: 3.18 | Melaka, Malaysia

Kolej Vokasional Besut 2022 – 2024
Diploma of Information Technology • CGPA: 3.55 • MUET: Band 3.5 | Terengganu, Malaysia

Kolej Vokasional Besut 2020 – 2022
Sijil Vokasional Malaysia — Computer & Network Services (Level 3) • CGPA: 3.52 | Terengganu, Malaysia

EXPERIENCE

Fakulti Teknologi Maklumat dan Komunikasi (FTMK), UTeM Jan 2025 – Present
Technical Associate (Rakan Teknikal) | Melaka, Malaysia

- Delivered rapid on-site hardware and software troubleshooting for 200+ faculty and student users across laboratory environments.
- Configured projectors, sound systems, and live-streaming equipment for 10+ hybrid faculty events with zero technical failures.
- Served as official event photographer and videographer; produced and edited promotional content for faculty social media channels.
- Supported faculty digital infrastructure including computer lab maintenance and network connectivity checks.

Port Klang Cruise Terminal Sdn Bhd Apr 2024 – Sep 2024
IT Intern | Port Klang, Selangor

- Executed routine system health checks and hardware/software verification across 50+ enterprise workstations, significantly reducing operational downtime.
- Assisted lead engineer in upgrading 20+ PoE switches and optimizing LAN/WAN infrastructure, improving overall network stability.
- Performed preventive maintenance and troubleshooting for 30+ CCTV cameras, ensuring 100% continuous surveillance coverage.
- Audited 200+ IT assets and implemented a lifecycle tracking system, achieving 100% documentation compliance.
- Prepared technical quotations for network components and evaluated vendor proposals for cost-effectiveness.

KEY PROJECTS

PanTalk — Privacy-First Messaging Platform (Current)

React, Node.js, Socket.IO, PostgreSQL, AWS

- Developing a secure real-time messaging platform with private messaging, temporary identities, QR-based connections, disappearing messages, group collaboration, media sharing, and privacy controls, deployed on AWS.

Threat Nexus XDR — Final Year Project (FYP): Cloud-Native Autonomous Security Response

AWS EC2, Lambda, DynamoDB, EventBridge, Flask, Chart.js

- Designed a cloud-native XDR platform combining a Cowrie SSH honeypot, a Lambda-based detection engine (10 attack rules), and automated IP blocking.
- Built real-time Telegram/SNS alerting and a Flask + Chart.js dashboard; validated detections via Kali Linux attack simulation.

Omniverse — Multi-Cloud Disaster Recovery (SRE Showcase)

Terraform, AWS EKS, GCP GKE, Python/Boto3, React & WebGL, Solidity

- Automated cross-cloud failover between AWS EKS and GCP GKE with Terraform-managed DR and a Python agent that reacts to SNS alerts to re-route traffic.
- Added chaos-engineering fault injection and an immutable Web3 audit trail logging failover events on-chain.

Cloud Security Posture Audit — Production-Grade CSPM Framework

Terraform, Prowler, Python, CIS Benchmarks, GitHub Actions

- Built a CIS-aligned CSPM framework combining Prowler with a custom compliance checker over Terraform-hardened infrastructure.
- Implemented CI/CD auto-remediation via GitHub Actions, lifting measured compliance from 42% to 87%.

Cloud Data Governance — Zero-Trust Engine for AWS S3 Data Lakes

Python, Terraform, AWS, Next.js

- Engineered auto-classification of PII into security tiers with redaction of sensitive data across S3 data lakes.
- Deployed honeytokens and automated quarantine workflows for compromised user accounts.

AEGIS — Cloud-Native SOC Dashboard

React, Three.js, AWS Amplify, Python (Boto3)

- Built an interactive 3D threat-globe visualization backed by AWS Amplify, with automated AWS remediation actions triggered via Boto3.

Niyyah — Islamic Super App for Muslim Malaysians

React, Next.js, Supabase, TailwindCSS

- Developed a production-grade fullstack app for mualaf (new converts) and warga emas (senior citizens): Al-Quran with audio, JAKIM-integrated prayer times, Muallaf Hub, and Zakat Calculator.

SmartChef — Serverless Recipe Recommendation Engine

AWS Lambda, API Gateway, DynamoDB, Python, Next.js

- Architected a serverless recipe engine on AWS with real-time market cost predictions and nutritional breakdowns via a Python backend.

FunTechPay — Neo-Banking Proof of Concept

React, FastAPI, DynamoDB, Terraform, JWT

- Built a 3D interactive banking dashboard with AI-driven spending insights, a split-bill QR engine, and JWT + TAC-secured transactions.

LEADERSHIP & INVOLVEMENT

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| • Gold Medalist — i-CPROM 2023 (UiTM Perlis), Intl. IT Innovation Competition | • Chairman — Mobile Legends Tournament, Volleyball Championship & iTech Club, KV Besut (2024) |
| • Protocol & Technical Committee — Program CEO Campus, UTeM (2025): VIP protocol & AV setups | • Technical Associate (Rakan Teknikal), FTMK UTeM (2025 – Present) |
| • Safety Secretariat, FTMK Programs (2025) | • Security Committee Member, Career Path Club (2025) |
| • Handball Representative, FTMK SAF Ke-19 (2025) | • Volunteer, TVET State & National Programs (2023 – 2025) |
| • Participant — Cloud Challenge DTC (2026) & UniPlay (2026) | |

CERTIFICATIONS & PROFESSIONAL TRAINING

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| • AWS Academy Cloud Security Builder (2026) | • AWS Academy Cloud Security Foundations (2026) |
| • Oracle Fusion AI Agent Studio Certified Foundations Associate (2026) | • Oracle Cloud Infrastructure 2025 Certified Foundations Associate (2026) |
| • Python Programming with Red Hat (AD141-RHA) (2026) | • AWS Academy Cloud Foundations (2025) |
| • Cisco Network Defense (2025) | • Junior Cybersecurity Analyst Career Path (2024) |
| • Endpoint Security (2024) | • CCNA: Introduction to Networks (2023) |
| • Ethical Hacker (2023) | • Cyber Threat Management (2023) |
| • Google Workspace Tools & Data Analysis (2025) | • RunCloud: Cloud Computing Fundamentals (2024) |
| • MyDIGITAL & Intel: AI Appreciate & Aware (2024) | • JPK: SKM Computer Network Services Level 3 (2023) |

SKILLS & INTERESTS

Cloud & Infrastructure: AWS (Lambda, API Gateway, DynamoDB, IAM, S3, EC2, EKS, EventBridge, GuardDuty, CloudTrail, Security Hub, Amplify), GCP (GKE), OpenStack, Docker, Terraform, Serverless, Virtualization

Security: Network Defense, Ethical Hacking, XDR, CSPM (Prowler, CIS Benchmarks), Zero-Trust Data Governance, Endpoint Security, Cyber Threat Management, Phishing Detection, SIEM/SOAR (foundational)

Networking: CCNA, LAN/WAN, PoE Switches, Load Balancing, VPN, Firewall Configuration, Intrusion Prevention

Programming: Python (Flask, FastAPI, Boto3), JavaScript/TypeScript, React, Next.js, C++, PHP, HTML/CSS, TailwindCSS, Chart.js, Three.js, Solidity, SQL, MySQL

IT Operations & Tools: Linux & Windows OS, Server Maintenance, CCTV Systems, IT Asset Auditing, AV Systems, Git, GitHub Actions, Docker, Supabase, RunCloud, Google Workspace, VS Code

Language / Interests: Malay (Native), English (Proficient – MUET Band 3.5) | Cloud security research, Islamic fintech, esports event management, photography & videography